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Winners share, followers gain: evidence from a national innovation award

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ABSTRACT

Knowledge spillovers are central to innovation policy, yet how honorary awards shape firms' knowledge sharing behavior remains underexplored. Using the China Patent Award and a sample of 24,236 firm-year observations from 2013 to 2023, this study finds that award-winning firms exhibit patent citation levels roughly 8% to 13% higher on average, with the increase concentrated in outward diffusion to other firms rather than internal knowledge recycling. The association is stronger under higher technological proximity and in technology-intensive industries, weakens amid intense market competition, and is amplified by robust intellectual property protection and greater media visibility. Collaborative patent activity and outward licensing following award receipt are both positively associated with subsequent knowledge diffusion, suggesting that public recognition attracts cooperative partners and creates licensing opportunities that broaden knowledge flows. Among non-winning firms, a one-unit increase in knowledge absorbed from award-winning firms is associated with roughly 9.2% more authorized invention patents. These findings suggest that honorary awards complement economic incentives by certifying technological leadership and reducing disclosure risks, generating social benefits that extend well beyond the originally recognized firm.

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Knowledge spillovers; innovation awards; patent citations; technological innovation

1. Introduction

Promoting knowledge spillovers is a central objective of innovation policy, yet firms tend to protect rather than share technological knowledge when incentives are inadequate. The social rate of return on R&D substantially exceeds the private rate (Lucking, Bloom, and Van Reenen 2019), and the non-rivalrous, partially non-excludable character of technological knowledge prevents firms from appropriating all returns from their R&D investments, suppressing voluntary disclosure. Where technological trajectories align closely, competitors can rapidly imitate leaders' innovations, further reinforcing defensive motivations. Governments have responded mainly with economic instruments such as R&D subsidies, tax incentives, and target-setting, which promote diffusion to varying degrees (He, Liu, and Ying 2023; Wang, Zheng, and Cao 2026). Far less attention has been paid to non-economic instruments, particularly honorary awards, which combine

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incentive and certification functions yet remain largely unexamined with respect to firms' knowledge spillover behavior.

Existing research offers several vantage points. One stream shows that intellectual property systems shape diffusion, with patent disclosure creating institutional conditions for knowledge flows while excessive protection may inhibit dissemination (Acs and Sanders 2012; De Rassenfosse, Pellegrino, and Raiteri 2024). Another documents positive externalities of government innovation policies for non-target firms and regions (He, Liu, and Ying 2023; Wang, Zheng, and Cao 2026). Work on the antecedents of spillovers has emphasized structural determinants such as geographic proximity, technological distance, and talent mobility (Battke et al. 2016; Drivas et al. 2020; Greunz 2005). A related strand examines prize-based and recognition mechanisms, finding that exploration-tolerant incentive structures expand recipients' willingness to disclose challenging work and that award receipt is associated with increased knowledge flows to peer firms (Azoulay, Graff Zivin, and Manso 2011; Kremer 1998; Manso 2011; You et al. 2024). What remains missing is a systematic account of whether honorary awards, as a specific form of government-conferred recognition combining incentive and certification, constitute a distinct channel for spillovers from winning firms, and whether such spillovers carry meaningful downstream consequences for non-winning firms' innovation. Spillovers are valuable, moreover, only insofar as recipient firms can absorb the knowledge and convert it into innovation output (Yaşar and Paul 2012), and this complete transmission process has not been tested in the context of award-driven spillovers.

This study therefore asks two questions. Can innovation honorary awards promote knowledge spillovers from award-winning firms? And if so, do these spillovers enhance the technological innovation of non-award-winning firms? Drawing on the China Patent Award, the study assembles a sample of Shanghai and Shenzhen A-share listed companies from 2013 to 2023 and estimates a two-way fixed effects model. Because the award is granted on merit rather than assigned by an external rule, the estimates are best understood as associations consistent with a certification account rather than as effects cleanly attributable to the award, and a layered set of checks is used to narrow competing explanations one by one.

The study makes three contributions. First, it introduces a non-economic incentive perspective into research on knowledge spillovers, showing that honorary awards are associated with the external diffusion of technological knowledge from winning firms and thereby extending the analytical scope of innovation incentive policy beyond conventional economic instruments. Second, it constructs a complete transmission chain from winners' knowledge spillovers to non-winners' technological innovation, extending the analytical perspective from individual firms to industry-level diffusion and clarifying the broader social benefits that recognition systems can generate. Third, by documenting how collaborative innovation, patent licensing, and media-amplified visibility shape the conditions under which recognition translates into knowledge sharing, it enriches understanding of the mechanisms linking honorary awards to diffusion and contributes emerging-economy evidence that may inform the design of similar recognition systems elsewhere.

2. Literature review and hypotheses

2.1. Literature review

Knowledge spillovers, the process through which technological knowledge crosses organizational boundaries, have long been central to innovation economics. Because R&D output is non-rivalrous and only partially excludable, firms cannot monopolize the value their innovation creates, and social returns systematically exceed private returns (Lucking, Bloom, and Van Reenen 2019). Patent citations have become the standard tool for tracing these flows, since a citation indicates that the cited patent's technological knowledge exerted substantive influence on the citing invention (Blazsek and Escribano 2010). Early work established citation-weighted patent counts as reliable indicators of innovation value and documented systematic geographic patterns in diffusion (Jaffe, Trajtenberg, and Henderson 1993; Trajtenberg 1990), and the robust relationship between citations and firm market value further validated the approach (Hall, Jaffe, and Trajtenberg 2005). Citation-based research has revealed the multidimensional character of spillovers. Spatially, diffusion decays with distance, as proximity reduces the friction costs of transmitting technical information (Greunz 2005; Perri et al. 2025). Technologically, the proximity between firms' knowledge bases governs absorption efficiency, and cross-field flows often require bridging technologies as intermediaries (Battke et al. 2016; Jee and Srivastav 2024). Organizationally, the mobility of skilled inventors carries tacit knowledge across firms (Drivas et al. 2020), executives with overseas work experience channel international knowledge into domestic firms (Feng et al. 2023), and universities and industry-university-research networks provide institutionalized conduits from academia to industry, with the spatial expansion of university campuses shown to stimulate surrounding innovation through collaboration and citation (Liu, Zhang, and Wang 2025; Quatraro and Scandura 2019; Yaşar and Paul 2012).

How fully spillover effects materialize depends on institutional environments and policy interventions. Stronger intellectual property protection reduces innovators' exposure to imitation and encourages codified public disclosure through patenting (De Rassenfosse, Pellegrino, and Raiteri 2024), though excessively strict protection can raise knowledge acquisition costs and slow subsequent innovation (Fabrizio 2007); the litigation process itself leaks technical details to competitors (Awate, Makhija, and Xiao 2024), and spillover risks abroad shape multinationals' patent enforcement decisions (Berry and Donnelly 2025). On the policy side, innovation target-setting promotes regional innovation and generates significant inter-city knowledge spillovers (Wang, Zheng, and Cao 2026), government-oriented support produces positive externalities for non-target firms that strengthen with geographic, cultural, and economic proximity (He, Liu, and Ying 2023), and increases in government R&D budgets accelerate cross-field diffusion between green and traditional technologies (Orsatti 2024).

A distinct but related body of work examines prize-based and recognition mechanisms. Patent buyouts have been proposed to decouple innovation incentives from monopoly rents (Kremer 1998); incentive designs that tolerate early failure and reward long-run exploration promote more novel innovation and shape how openly inventions are shared (Manso 2011); recognition-based funding is linked to higher-impact and more original research, consistent with awards easing career concerns and expanding willingness to disclose challenging work (Azoulay, Graff Zivin, and Manso 2011); and award receipt has been associated with increased knowledge flows to peer firms and downstream effects on recipients' innovation output (You et al. 2024). The present study extends this line by focusing specifically on the China Patent Award, decomposing spillovers into external and internal dimensions to identify the pattern driving citation increases, examining how market competition, intellectual property protection, and media visibility condition diffusion, conducting a within-firm patent-level comparison between awarded and non-awarded patents to address quality confounding, testing patent licensing alongside collaboration as transmission channels, and tracing the transmission chain through to the authorized invention patents of non-winning firms.

Against this backdrop, a gap emerges. The policy-oriented strand of this literature has concentrated on economic instruments such as subsidies, tax incentives, and target-setting, and although prize-based recognition has begun to attract attention, whether honorary awards can serve as a distinct channel for spillovers from winning firms, and whether those spillovers matter for the innovation trajectories of non-winning firms, has not been directly addressed. This study takes the China Patent Award as its research object to examine these questions.

2.2. Hypothesis development

Knowledge spillovers refer to technological knowledge generated in firms' R&D activities being acquired at low or no cost and utilized by other market participants (Aghion and Jaravel 2015). Follower firms can innovate incrementally on the achievements of frontier firms, avoiding duplicative R&D investment and accelerating industry-wide technological progress (Gyamfi, Agozie, and Bekun 2022). In practice, however, spillovers face significant barriers. Because technological knowledge is non-rivalrous and only partially excludable, private R&D returns fall systematically below social returns, suppressing voluntary disclosure (Chen, Gavious, and Lev 2017; He and Lee 2023). Where technological trajectories align closely, competitors can convert imitated innovations into market share, and this business stealing effect erodes innovators' expected profits and reinforces defensive motivations (Operti and Carnabuci 2014; Seo and Sonn 2019). Designing instruments that stimulate sharing while preserving innovation incentives is therefore a core problem in innovation policy.

Innovation honorary awards, as a non-economic instrument combining incentive and certification functions, can loosen these barriers through three pathways. First, a risk control function. An award conferred by national authoritative institutions under strict standards credibly signals to the market that the winner leads in its technological field. This official certification strengthens the institutional protection of the winner's intellectual property, raising potential infringers' violation costs and detection probability while providing the firm stronger grounds for rights protection, thereby reducing the commercial risk of disclosure. When firms are confident that their technological lead is institutionally protected rather than easily eroded, their reluctance to disclose diminishes. Second, a resource aggregation function. The China Patent Award has strict selection criteria and limited quotas, and this scarcity demonstrates winners' innovation strength and potential, attracting talent, capital, and cooperation opportunities. Beyond lowering search and matching costs, when firms connect with external entities through strategic alliances, collaborative R&D, and technology licensing, knowledge spillovers become a natural byproduct of cooperative relationships rather than a unilateral loss (Audretsch, Belitski, and Caiazza 2025; Uzzi and Gillespie 2002). Third, a reputation maintenance function. Government-conferred awards amplified by mainstream media rapidly raise winners' visibility and credibility, yet reputation is an intangible asset requiring continuous behavioral investment. To consolidate their social image as industry leaders, winners have stronger motivation to demonstrate technological leadership through technology sharing, participation in standard-setting, and industry-university-research collaboration, and these behaviors objectively broaden and deepen spillovers (Zhang, Yu, and Xia 2014). Accordingly, this paper proposes:

H1: Innovation honorary awards are positively associated with the knowledge spillover level of award-winning firms.

The ultimate value of spillovers depends on whether diffused knowledge is effectively absorbed and converted into innovation output by other firms. Absorptive capacity theory holds that the benefit a firm derives from external knowledge depends on its ability to recognize, digest, and commercially apply that knowledge, which rests on prior accumulation of related knowledge bases (Camisón and Forés 2010; Easterby-Smith et al. 2008). Honorary awards facilitate non-winners' absorption by releasing two kinds of signals. One is a strategic direction signal. The China Patent Award's criteria span patent quality, technological advancement, and economic and social benefits, so winning achievements mark the intersection of national industrial policy orientation and market demand trends. For non-winners, this public information offers a credible reference for reading technological frontiers and anticipating market directions, reducing the costs of searching for technical information and discerning investment directions under uncertainty and freeing resources for substantive innovation (McGee and Sawyerr 2003; Sabherwal et al. 2019). The other is an innovation quality signal. Achievements emerging from rigorous review typically represent the technological frontier, so knowledge spilling from winners carries high quality and application value. Non-winners that learn from this market-validated knowledge through patent citations, technical literature tracking, and industry exchange can avoid common pitfalls in R&D and reduce trial-and-error costs and failure risks (Rui, Cuervo-Cazurra, and Un 2016; Stojcic, Vujanovic, and Radosevic 2026). Clear technology benchmarks established by awards also sharpen the competitive reference system facing non-winners, generating catch-up pressure alongside learning direction and incentivizing investment in absorption and transformation capabilities. Accordingly, this paper proposes:

H2: Knowledge spillovers from award-winning firms are positively associated with the technological innovation level of non-award-winning firms.

3. Methodology

3.1. Sample and data

This study integrates multiple data sources. Patent application, authorization, citation detail, and patent-level characteristic data come from the National Intellectual Property Administration (CNIPA) and the PATSNAP patent database; listed company financial and governance data from the CSMAR database; China Patent Award information from CNIPA official announcements; patent license and assignment records from CNIPA legal-status data and PATSNAP transaction records; regional intellectual property protection data from the China Marketization Index compiled by the National Economic Research Institute (NERI); and media coverage data from the CNRDS database. Matching proceeded in three steps. Winning patents from the 11th through 25th China Patent Awards were retrieved by patent number to build a winning-patent database; the 'current patent holder' field in citation details was matched to the listed company roster to link cited and citing parties and construct a firm-level citation network; and patent-level citation records were aggregated to the firm-year level to compute each firm's annual knowledge spillover and inflow indicators. Considering that the selection cycle was once every two years before the 11th edition and became annual from the 11th edition onward, to reduce the influence of award timing intervals this paper selects Shanghai and Shenzhen A-share listed companies from 2013 to 2023 as the initial sample and excludes firms in service industries such as finance, insurance, and real estate; ST, *ST, and PT firms; observations with missing key variables or abnormal financial data; and firms with no patent application activity or R&D expenditure during the sample period. All continuous variables are winsorized at the top and bottom 1%. The final unbalanced panel contains 24,236 firm-year observations. In addition, a patent-year panel of 138,246 observations, restricted to firms with at least one award-winning patent, supports the within-firm patent-level analysis.

3.2. Variables

This study operationalizes knowledge spillovers as codified flows traceable through patent citation linkages. Citations provide an observable and replicable measure of inter-firm knowledge flows, since each citation indicates that the citing patent drew on the cited patent's technological knowledge (Blazsek and Escribano 2010). The approach does not capture tacit transmission through personnel mobility or informal exchange, but it precisely tracks the direction, timing, and intensity of flows at the firm level and is consistent with a substantial empirical literature (Battke et al. 2016; Hall, Jaffe, and Trajtenberg 2005; Jee and Srivastav 2024).

The dependent variable is knowledge spillover level (SPILLOVER), the natural logarithm of one plus the total citations received in a year by all of a firm's invention and utility model patents; the logarithmic transformation addresses the right-skewed distribution of citation counts and retains zero-citation observations, a standard practice (Lei and Xu 2025). In robustness analyses this aggregate is decomposed into external spillovers (SPILLOVEROUT, citations received from other firms) and internal spillovers (SPILLOVERSELF, self-citations), and the year-on-year change ΔSPILLOVER is used to test whether the award is associated with an acceleration in citation growth rather than merely a higher level.

The core explanatory variable is innovation honorary award (AWARD), equal to one from the year a firm first receives the China Patent Award onward and zero otherwise. Given that invention and utility model patents involve deeper technological improvement, the study focuses on firms receiving Gold, Silver, and Excellence Awards, identifying multi-award firms by their first award year. This absorbing treatment design reflects the expectation that the signaling and certification effects of a national-level award persist beyond the award year, an expectation the event-study evidence in the results section supports.

Control variables cover seven aspects: firm size (SIZE, the natural logarithm of the number of employees), asset-liability ratio (LEV, total liabilities over total assets), return on assets (ROA, net profit over average total assets), firm growth (GROWTH, operating revenue growth rate), ownership nature (SOE, one if the actual controller is a government entity), R&D intensity (RD, annual R&D expenditure over total assets), and firm age (AGE, the natural logarithm of years since IPO).

To address the concern that high-quality patents may independently drive both award receipt and citation accumulation, two firm-year patent quality proxies are constructed: the average number of claims per patent (AVGCLAIMS), reflecting the breadth of protection scope, and the average number of backward citations per patent (AVGBCITE), measuring the depth of prior-art engagement; both are averaged across all invention and utility model patents held by a firm in a year and included in extended baseline specifications. To capture the depth of a firm's innovation history, an accumulated patent stock variable (PATENTSTOCK) is defined as the natural logarithm of the cumulative count of a firm's granted patents over the five years preceding each observation, excluding any patents granted in or after the award year, so that the measure does not absorb patents that the award itself may have helped generate. A pre-award citation baseline (PRECITATION), defined as the natural logarithm of one plus the firm's mean annual patent citations over the three years preceding the award year, addresses the concern that post-award increases may continue a pre-existing trajectory.

For the mechanism and boundary condition analyses, three further variables are constructed: collaborative patents (COPATENT), the natural logarithm of one plus the number of patents jointly filed with at least one external organization in a year; outward licensing (LICENSEOUT), the natural logarithm of one plus the number of the firm's patents licensed or transferred to external parties in a year, built from the license and assignment records described above; and media attention (MEDIA), the natural logarithm of one plus the number of news reports mentioning the firm in a year, from the CNRDS database.

3.3. Empirical strategy

To test whether innovation honorary awards are associated with higher knowledge disclosure by award-winning firms, this study estimates the following two-way fixed effects model:

$$\mathrm{SPILLOVER}_{i,t} = \beta_0 + \beta_1 \mathrm{AWARD}_{i,t-1} + \beta_2^{'} \mathrm{CONTROLS}_{i,t-1} + \gamma_i + \delta_t + \varepsilon_{i,t}$$

where subscripts *i* and *t* denote firm and year, $\mathrm{SPILLOVER}_{i,t}$ is firm *i* 's knowledge spillover level in year t, $\mathrm{AWARD}_{i,t-1}$ indicates whether the firm was in an award-winning state in the prior year, $\mathrm{CONTROLS}_{i,t-1}$ is the control variable vector, γ_i and δ_t are firm and year fixed effects, and $\varepsilon_{i,t}$ is the error term. All explanatory and control variables are lagged by one period to mitigate reverse causality, with standard errors clustered at the firm level. In extended specifications, AVGCLAIMS, AVGBCITE, PATENTSTOCK, and PRECITATION are progressively added to the control vector.

The model for testing the innovation effects on non-winning firms takes the form:

$$\mathrm{INNOVATION}_{i,t} = \beta_0 + \beta_1 \mathrm{SPILLOVERIN}_{i,t-1} + \beta_2^{'} \mathrm{CONTROLS}_{i,t-1} + \gamma_i + \delta_t + \varepsilon_{i,t}$$

where $\mathrm{INNOVATION}_{i,t}$ is the natural logarithm of one plus non-award-winning firm i 's authorized invention patents in year t, and $\mathrm{SPILLOVERIN}_{i,t-1}$ is the natural logarithm of one plus the firm's citations to award-winning firms' patents in year $t-1$. The regression sample for this model includes only firms that never received the award during the sample period.

3.4. Scope of the empirical approach

The empirical strategy estimates associations under a parallel-trends assumption, namely that award-winning and comparison firms would have followed similar citation trajectories absent the award, an assumption the event-study evidence supports. The merit-based nature of the award shapes what this approach can deliver. A sharp regression-discontinuity design is unavailable, because the China Patent Award is conferred through expert review whose scores are not disclosed and whose grades are allocated by ranking under limited quotas rather than an absolute threshold, leaving no public discontinuity to exploit. An instrumental-variable approach is likewise difficult: the most plausible source of variation in the probability of winning, differences across provinces and industry associations in recommendation quotas and competition, is itself tied to local innovation capacity and thus bears directly on knowledge diffusion, so the exclusion requirement fails; more generally, because the award rewards merit, variables associated with a higher probability of winning tend also to signal greater underlying innovativeness, the very confound an instrument would need to avoid. The most natural quasi-experimental route would compare awarded patents against patents nominated but not ultimately selected, since such near-miss candidates clear the same recommendation screen and closely resemble winners in unobserved quality. The obstacle is data availability rather than logic: CNIPA publishes only the final recipient list, no consolidated national registry of nominated-but-not-awarded patents exists, and the provincial notices that do appear differ in coverage and are frequently withdrawn after their comment windows close, so a reliable full-sample comparison is not feasible with currently accessible records and is set aside for subsequent research. Given these constraints, this study positions its central finding as a robust association consistent with a certification account, supported by a layered set of checks that progressively narrows competing explanations, from firm-level controls, patent quality proxies, accumulated patent stock, and propensity-score matching, through event-study and placebo tests and a staggered-adoption estimator, to a within-firm comparison of awarded and non-awarded patents, with residual selection acknowledged where it arises.

4. Results

4.1. Summary statistics

Table 1 reports descriptive statistics. SPILLOVER has a mean of 1.486 and a standard deviation of 1.524, indicating considerable variation in citation outcomes; AWARD has a mean

Table 1. Descriptive statistics.

Variable	N	Mean	Std. Dev.	Min	Max
SPILLOVER	24,236	1.486	1.524	0.000	5.872
AWARD	24,236	0.118	0.323	0.000	1.000
SIZE	24,236	7.582	1.186	4.654	11.024
LEV	24,236	0.412	0.196	0.052	0.878
ROA	24,236	0.038	0.062	−0.228	0.196
GROWTH	24,236	0.156	0.348	−0.486	2.124
SOE	24,236	0.312	0.463	0.000	1.000
RD	24,236	0.046	0.038	0.000	0.178
AGE	24,236	2.682	0.612	0.693	3.892
AVGCLAIMS	24,236	8.460	4.820	1.000	28.640
AVGBCITE	24,236	6.240	3.680	0.000	22.480
COPATENT	24,236	0.384	0.612	0.000	3.218
LICENSEOUT	24,236	0.216	0.484	0.000	2.708
MEDIA	24,236	3.842	1.264	0.000	7.486

of 0.118, consistent with the scarcity of the China Patent Award; RD averages 4.6% of total assets. The patent quality proxies show substantial cross-firm heterogeneity (AVGCLAIMS mean 8.46, AVGBCITE mean 6.24), and the COPATENT and LICENSEOUT means of 0.384 and 0.216 indicate that collaboration and licensing activity are concentrated among a subset of firms. Table 2 shows that award-winning firms exhibit significantly higher SPILLOVER (2.347 versus 1.372, significant at the 1% level), together with higher RD, AGE, AVGCLAIMS, AVGBCITE, COPATENT, and LICENSEOUT, confirming that winners tend to hold higher-quality patent portfolios and underscoring the need to control for patent quality at both firm and patent levels. Correlation coefficients among the main variables are all below 0.5 in absolute value, and the maximum variance inflation factor is 2.48, indicating no serious multicollinearity.

4.2. Main results

Table 3 reports the baseline results. Column (1), with only firm and year fixed effects, yields an AWARD coefficient of 0.152, significant at the 1% level. Adding the full control set in Column (2) leaves the coefficient at 0.134, still at the 1% level, so the positive tie between the award and knowledge spillovers survives adjustment for firm size, innovation investment, and maturity. Column (3) adds the two patent quality proxies; the

Table 2. Group mean comparison.

Variable	Award-winning Firms	Non-award-winning Firms	Difference	t-value
SPILLOVER	2.347	1.372	0.975***	18.64
SIZE	8.124	7.509	0.615***	15.23
LEV	0.428	0.410	0.018**	2.56
ROA	0.046	0.037	0.009***	4.12
GROWTH	0.178	0.153	0.025**	2.08
SOE	0.386	0.302	0.084***	5.34
RD	0.058	0.044	0.014***	4.86
AGE	2.814	2.664	0.150***	6.42
AVGCLAIMS	10.240	8.180	2.060***	6.84
AVGBCITE	7.860	6.020	1.840***	7.12
COPATENT	0.582	0.348	0.234***	7.46
LICENSEOUT	0.348	0.192	0.156***	6.24

Note: ***, **, * denote significance at the 1%, 5%, 10% levels respectively.

Table 3. Baseline results.

Variable	(1) SPILLOVER	(2) SPILLOVER	(3) SPILLOVER	(4) SPILLOVER
AWARD	0.152***	0.134***	0.122***	0.076**
	(3.45)	(3.18)	(2.76)	(2.04)
PATENTSTOCK				0.214***
				(17.62)
AVGCLAIMS			0.024*	0.020
			(1.86)	(1.58)
AVGBCITE			0.032**	0.026*
			(2.14)	(1.82)
SIZE		0.248***	0.238***	0.192***
		(11.84)	(11.36)	(8.86)
LEV		0.314***	0.308***	0.286***
		(4.18)	(4.08)	(3.78)
ROA		0.278***	0.274***	0.262***
		(3.56)	(3.48)	(3.34)
GROWTH		−0.022*	−0.020*	−0.018
		(−1.84)	(−1.76)	(−1.56)
SOE		0.014	0.012	0.010
		(0.34)	(0.28)	(0.24)
RD		0.486***	0.472***	0.408***
		(4.24)	(4.10)	(3.56)
AGE		0.024**	0.022**	0.016
		(2.06)	(1.98)	(1.42)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	24,236	24,236	24,236	24,236
Adj. R^2	0.392	0.412	0.424	0.448

Note: t-values in parentheses; ***, **, * denote significance at the 1%, 5%, 10% levels respectively. PATENTSTOCK is the accumulated count of granted patents over the five years preceding each observation, excluding patents granted in or after the award year.

AWARD coefficient settles at 0.122, again at the 1% level, with AVGBCITE entering positively, as patents built on a richer prior-art foundation sit more centrally in citation networks, and AVGCLAIMS entering only marginally, reflecting that claim breadth is a noisier quality signal at the firm-year level.

Column (4) brings accumulated prior patent stock into the baseline itself, following the view that a firm's innovation history belongs in the main specification rather than only in supplementary checks. The AWARD coefficient attenuates to 0.076, remaining significant at the 5% level, while PATENTSTOCK enters strongly at the 1% level. Because a firm's own later patenting can respond to the award, conditioning on patent stock removes part of the association it is meant to isolate, so 0.076 is best taken as a conservative lower bound. Together, the quality-adjusted 0.122 and the stock-adjusted 0.076 bracket the association and imply that award-winning firms are tied to patent citation levels roughly 8% to 13% higher on average, a range reported in preference to a single point estimate so as neither to overstate nor to understate the magnitude, and both bounds support H1. The quality-adjusted specification serves as the reference for the heterogeneity and mechanism analyses that follow, since patent stock, being responsive to the award, is better left out of those decompositions.

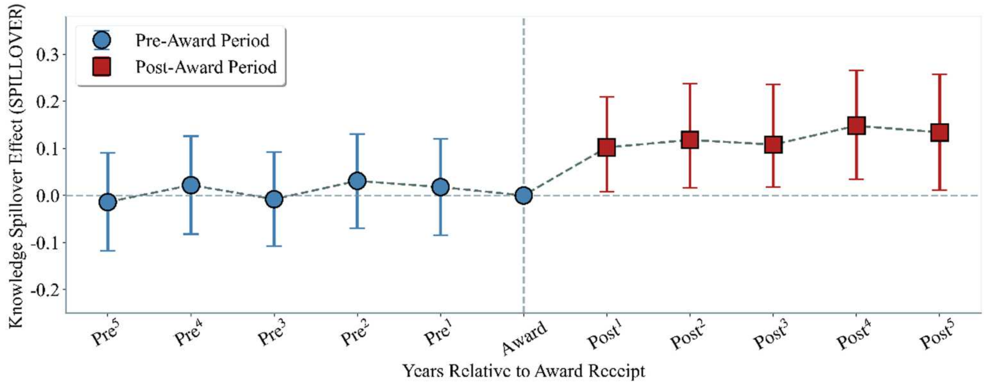

Figure 1. Parallel trends test (citation level).

4.3. Robustness checks

An event study first tests the parallel trends assumption. Figure 1 displays dynamic estimates for five periods before and after award receipt. Pre-treatment coefficients are small in magnitude and statistically insignificant throughout, suggesting that award-winning and non-award-winning firms followed similar spillover trajectories prior to the event; post-award coefficients turn significantly positive from the first post-award year onward and remain persistently elevated throughout the window, with natural period-to-period variation but no reversion toward zero, consistent with a sustained rather than transitory association.

A related concern is that the post-award citation increase might reflect a nonlinear continuation of a pre-existing quality-driven growth trajectory. Figure 2 therefore repeats the event study with ΔSPILLOVER as the dependent variable. Pre-award coefficients are consistently indistinguishable from zero, indicating that eventual winners did not exhibit systematically faster citation growth before the award relative to the comparison group, while post-award coefficients turn positive and reach significance at the 5% level from year one onward, consistent with a modest but persistent acceleration in citation growth rates.

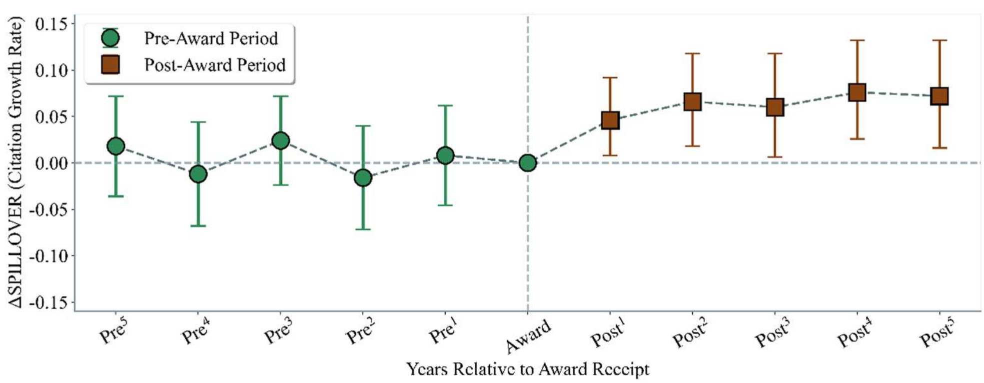

Figure 2. Event study using citation growth rate (ΔSPILLOVER).

Decomposing the aggregate measure clarifies what kind of diffusion drives the baseline. Table 4 shows that the AWARD association with external citations is significant (0.136, $p < 0.01$) while the association with self-citations is not (0.022, $p > 0.10$). This pattern confirms that the baseline result reflects genuine cross-organizational knowledge diffusion rather than a general increase in internal patent activity, and that the aggregate SPILLOVER measure is a reasonable indicator of inter-firm knowledge flows.

A further battery of checks, reported in Appendix A, corroborates the baseline. Propensity score matching on all controls yields a matched-sample AWARD coefficient of 0.118 ($p < 0.01$); a system GMM specification with a lagged dependent variable gives 0.098 ($p < 0.05$) with valid instruments; the Callaway and Sant'Anna (2021) staggered-adoption estimator returns an aggregated ATT of 0.112 ($p < 0.05$), addressing heterogeneity bias in the two-way fixed effects estimator under staggered timing (Callaway and Sant'Anna 2021); a placebo exercise randomly reassigning award timing 500 times centers the coefficient distribution on zero with the actual estimate falling clearly in the right tail; controlling for the pre-award citation baseline PRECITATION leaves an AWARD coefficient of 0.094 ($p < 0.05$); and using ΔSPILLOVER as the dependent variable yields 0.064 ($p < 0.05$), echoing Figure 2.

4.4. Within-firm patent-level analysis

The checks in Section 4.3 address the quality confound at the firm level. A more direct approach shifts the unit of analysis to the individual patent and compares, within the same firm, the citation trajectories of award-winning and non-awarded patents over the same period. Because both groups belong to the same firm, all firm-level attributes are held constant, and differences in post-award citation dynamics are attributable to the award status of the specific patent rather than to firm-level heterogeneity. It is useful to state precisely what this comparison delivers. Holding the firm fixed removes time-invariant firm-level confounders, the specific concern raised about the firm-level results, yet it does not amount to random assignment of the award across otherwise identical patents, since awarded patents are selected on merit and may still differ from their non-awarded counterparts along dimensions that patent-level controls do not fully capture. For a merit-based award, a within-patent randomization is not attainable. This analysis should therefore be understood as removing one important and specific source of confounding rather than as delivering a clean attribution, and the residual patent-level selection is acknowledged in the closing discussion.

Table 4. Decomposition of knowledge spillovers.

Variable	(1) SPILLOVEROUT	(2) SPILLOVERSELF
AWARD	0.136***	0.022
	(3.28)	(0.52)
Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
N	24,236	24,236
Adj. R^2	0.412	0.424

Note: t-values in parentheses; ***, **, * denote significance at the 1%, 5%, 10% levels respectively.

The patent-year sample comprises 138,246 observations from firms holding at least one award-winning patent, spanning 2013–2023. The within-firm model is specified as:

$$CITATION_{p,t} = \beta_0 + \beta_1(AWARDED_p \times POST_t) + \beta_2 POST_t + \gamma' X_{p,t} + \eta_p + \delta_t + \varepsilon_{p,t}$$

where $CITATION_{p,t}$ is the natural logarithm of one plus citations received by patent p in year t, $AWARDED_p$ equals one if the patent received the China Patent Award, and $POST_t$ equals one from the award year onward. The vector $X_{p,t}$ contains four patent-level controls: the log of patent age (years since application date plus one), capturing mechanical citation accumulation with maturity; the log of one plus claims count, proxying protection breadth; the log of one plus backward citation count, proxying technological rootedness in prior art; and a patent type indicator equal to one for invention patents and zero for utility model patents. Patent fixed effects η_p absorb all time-invariant patent characteristics including intrinsic technological quality, year fixed effects δ_t are included throughout, and an extended specification adds year-by-IPC-class interactions to control for field-specific citation trends.

Table 5 reports three specifications with progressively richer controls. With only patent and year fixed effects, the AWARDED × POST coefficient is 0.078 ($p < 0.05$), an implied incremental citation gain of approximately 8.1% for awarded patents relative to non-awarded patents from the same firm after the award is conferred. Adding patent age and four-digit IPC class fixed effects lowers the coefficient slightly to 0.072, ruling out composition effects of patent vintage or technology field within the firm, and the full specification with the remaining patent-level controls and year-by-IPC interactions yields 0.068, still significant at the 5% level. Patent age enters at about 0.178, consistent with citation accumulation over time; claims and backward citations carry modest positive coefficients, reflecting that broader and more technically grounded patents attract more attention; and invention patents receive more citations than utility model patents. The patent-level magnitude of roughly 0.068–0.078 is smaller than the firm-level quality-adjusted 0.122, as expected: the firm-level estimate aggregates citation

Table 5. Patent-level within-firm analysis.

Variable	(1) CITATION	(2) CITATION	(3) CITATION
AWARDED × POST	0.078**	0.072**	0.068**
	(2.24)	(2.18)	(2.06)
POST	0.042**	0.038*	0.034*
	(2.08)	(1.92)	(1.72)
Ln (AGE + 1)		0.186***	0.178***
		(12.46)	(11.82)
Ln (CLAIMS + 1)			0.042**
			(2.18)
Ln (BCITE + 1)			0.028*
			(1.84)
TYPE			0.064**
			(2.06)
Patent FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
IPC Class FE	No	Yes	Yes
Year × IPC FE	No	No	Yes
N	138,246	138,246	138,246
Adj. R^2	0.312	0.328	0.346

Note: t-values in parentheses; ***, **, * denote significance at the 1%, 5%, 10% levels respectively.

gains across the entire portfolio while the patent-level estimate isolates the increment specific to the awarded patents. Evidence at both levels points in the same direction.

4.5. Heterogeneity and boundary conditions

The baseline association may vary with the technological environment. Table 6 examines two dimensions. The first is technological proximity (PROXIMITY), measured by the mean Jaccard similarity between a firm's four-digit IPC codes and those of same-industry firms, with higher values indicating a technological base closer to industry peers. Splitting the sample at the year-industry median, the AWARD coefficient is 0.164 ($p < 0.01$) in the high-proximity subsample versus 0.082 ($p < 0.10$) in the low-proximity subsample, with the between-group difference significant at the 5% level: common knowledge backgrounds lower the cognitive barriers to cross-organizational learning. The second is industry technology intensity (TECHINTENSITY), the ratio of total industry R&D expenditure to operating revenue, with the sample split by the median of industry means. The AWARD coefficient is 0.158 ($p < 0.01$) in high-technology industries and an insignificant 0.074 in low-technology industries, the difference again significant at the 5% level. In high-technology industries, faster technology turnover, denser citation networks, and more codified knowledge make award signals easier for market participants to capture and utilize; in traditional industries, progress rests more on experience accumulation and tacit knowledge, limiting patent citations as a diffusion channel.

Table 7 explores three external boundary conditions. The first is market competition (COMPETITION), measured by the inverse of the Herfindahl index. The AWARD coefficient in Column (1) is 0.184 ($p < 0.01$) at average competition levels, and the AWARD × COMPETITION interaction is −0.448 ($p < 0.05$), indicating that more intense competition weakens the association: when competitive pressure is high, even winners protect core technologies rather than disclose them to potential rivals, reflecting the strategic trade-off between knowledge sharing and market position. The second is intellectual property protection (IPR), measured by the NERI China Marketization Index sub-index. The AWARD coefficient is 0.074 ($p < 0.10$) and the AWARD × IPR interaction 0.064 ($p < 0.05$), indicating that stronger regional protection amplifies the association: when legal frameworks let firms extract returns from disclosed knowledge through licensing and cooperative arrangements, the risk of sharing falls and voluntary disclosure becomes more attractive. The third is media visibility. The reputation maintenance channel in Section 2.2 implies that more visible firms face stronger reputational incentives to demonstrate technological

Table 6. Cross-sectional analysis.

Variable	(1) SPILLOVER High PROXIMITY	(2) SPILLOVER Low PROXIMITY	(3) SPILLOVER High TECHINTENSITY	(4) SPILLOVER Low TECHINTENSITY
AWARD	0.164***	0.082*	0.158***	0.074
	(3.08)	(1.68)	(3.04)	(1.40)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	9,486	14,750	8,924	15,312
Adj. R^2	0.420	0.392	0.414	0.384

Note: t-values in parentheses; ***, **, * denote significance at the 1%, 5%, 10% levels respectively.

Table 7. Boundary conditions.

Variable	(1) SPILLOVER	(2) SPILLOVER	(3) SPILLOVER
AWARD	0.184***	0.074*	0.082*
	(3.58)	(1.92)	(1.88)
COMPETITION	0.158*		
	(1.94)		
AWARD × COMPETITION	−0.448**		
	(−2.08)		
IPR		0.028*	
		(1.76)	
AWARD × IPR		0.064**	
		(2.08)	
MEDIAHIGH			0.124***
			(4.62)
AWARD × MEDIAHIGH			0.078**
			(2.12)
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	24,236	24,236	24,236
Adj. R^2	0.416	0.418	0.426

Note: t-values in parentheses; ***, **, * denote significance at the 1%, 5%, 10% levels respectively.

leadership through sharing, as both the rewards of visible sharing and the costs of perceived protectionism are amplified under media scrutiny. MEDIAHIGH equals one for firm-years with above-median annual media coverage. In Column (3), the AWARD coefficient among low-media firms is 0.082 ($p < 0.10$) and the AWARD × MEDIAHIGH interaction is 0.078 ($p < 0.05$), indicating a substantially stronger association among firms receiving greater media attention; the MEDIAHIGH main effect of 0.124 ($p < 0.01$) confirms that higher-visibility firms diffuse more broadly regardless of award status, with the interaction capturing the incremental amplification tied specifically to recognition.

4.6. Collaboration and licensing as transmission channels

This section considers two channels through which award recognition may connect to higher spillovers: expanded collaborative innovation and outward patent licensing. The resource aggregation function developed in the hypotheses implies that public certification of technological leadership both attracts cooperative partners and opens opportunities to license technology outward, and that co-innovation and licensing each carry knowledge across firm boundaries as a byproduct. Column (1) of Table 8 restates the reference association from the quality-adjusted specification, in which the AWARD coefficient on SPILLOVER is 0.122. Columns (2) and (4) relate award receipt to each channel in turn: award receipt is followed by higher collaborative patent activity (0.096, $p < 0.01$) and greater outward licensing (0.058, $p < 0.05$), patterns consistent with recognition lowering the search and trust frictions that ordinarily slow cooperation and technology transactions. Columns (3), (5), and (6) reintroduce the channel variables into the spillover equation, first singly and then jointly. With both present in Column (6), the AWARD coefficient falls to 0.098 while COPATENT enters at 0.166 and LICENSEOUT at 0.138, so part of the total association travels through the two cooperative channels while the larger part remains with the direct award term. Bootstrap procedures with 1,000 replications place

Table 8. Collaboration and licensing channels.

Variable	(1) SPILLOVER	(2) COPATENT	(3) SPILLOVER	(4) LICENSEOUT	(5) SPILLOVER	(6) SPILLOVER
AWARD	0.122***	0.096***	0.104**	0.058**	0.113**	0.098**
	(2.76)	(3.04)	(2.42)	(2.45)	(2.56)	(2.28)
COPATENT			0.186***			0.166***
			(3.02)			(2.76)
LICENSEOUT					0.155***	0.138***
					(2.90)	(2.62)
AVGCLAIMS	0.024*	0.018	0.022*	0.012	0.022*	0.020*
	(1.86)	(1.42)	(1.78)	(1.02)	(1.80)	(1.66)
AVGBCITE	0.032**	0.024*	0.028*	0.016	0.028*	0.026*
	(2.14)	(1.78)	(1.86)	(1.28)	(1.84)	(1.74)
SIZE	0.238***	0.208***	0.236***	0.156***	0.234***	0.230***
	(11.36)	(9.24)	(11.28)	(7.18)	(11.20)	(11.14)
LEV	0.308***	0.182*	0.304***	0.128	0.306***	0.300***
	(4.08)	(1.82)	(4.02)	(1.52)	(4.04)	(3.96)
ROA	0.274***	0.218**	0.270***	0.142*	0.272***	0.264***
	(3.48)	(2.56)	(3.42)	(1.78)	(3.44)	(3.36)
GROWTH	−0.020*	−0.012	−0.018*	−0.006	−0.018*	−0.016
	(−1.76)	(−0.96)	(−1.68)	(−0.58)	(−1.66)	(−1.48)
SOE	0.012	0.026	0.010	0.020	0.010	0.008
	(0.28)	(0.62)	(0.26)	(0.52)	(0.26)	(0.22)
RD	0.472***	0.516***	0.468***	0.308***	0.466***	0.458***
	(4.10)	(4.04)	(4.06)	(3.14)	(4.04)	(4.02)
AGE	0.022**	0.016	0.020**	0.010	0.020*	0.018*
	(1.98)	(1.36)	(1.96)	(0.92)	(1.90)	(1.72)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	24,236	24,236	24,236	24,236	24,236	24,236
Adj. R^2	0.424	0.384	0.432	0.312	0.430	0.438

Note: t-values in parentheses; ***, **, * denote significance at the 1%, 5%, 10% levels respectively.

the collaboration channel at 0.016 (95% confidence interval 0.003–0.030) and the licensing channel at 0.008 (95% confidence interval 0.001–0.017), each excluding zero at the 5% level. Measured against the reference of 0.122, collaboration accounts for roughly 13% and licensing for roughly 7%, so the two observed channels together carry about one fifth of the association. They are contributing rather than exhaustive channels; the remaining share reflects channels the present data cannot separately observe, among them voluntary disclosure through non-patent means, participation in standard-setting, and the movement of inventors across organizational boundaries. Among the controls, RD and SIZE remain positive and significant, as larger and more research-intensive firms cooperate and license more actively regardless of award status.

4.7. Effects on non-winners

Knowledge spillovers are not the ultimate policy objective; their value lies in whether they can be effectively absorbed by other firms and translated into innovation output. This section tests that downstream question on non-award-winning firms, with results in Table 9. One caveat deserves emphasis before the estimates. The knowledge inflow measure reflects a firm's own citation behavior, which is itself a choice: firms inclined to draw on external frontier knowledge may also be more innovative for reasons unrelated to the award, and more innovative firms may cite more prolifically, so the direction of association cannot be settled by the data alone. The estimates in this section are

Table 9. Innovation demonstration effects.

Variable	(1) INNOVATION	(2) INNOVATION	(3) INNOVATION
SPILLOVERIN	0.112***	0.088***	0.082***
	(9.28)	(7.96)	(7.42)
PROXIMITYW			0.124**
			(2.36)
SIZE		0.286***	0.284***
		(11.86)	(11.74)
LEV		0.162**	0.158*
		(1.98)	(1.94)
ROA		0.374***	0.372***
		(4.38)	(4.36)
GROWTH		0.038***	0.038***
		(2.74)	(2.76)
SOE		−0.024	−0.022
		(−0.62)	(−0.58)
RD		0.542***	0.538***
		(4.62)	(4.58)
AGE		0.028*	0.026*
		(1.88)	(1.82)
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	21,408	21,408	21,408
Adj. R^2	0.406	0.418	0.422

Note: t-values in parentheses; ***, **, * denote significance at the 1%, 5%, 10% levels respectively.

accordingly more exposed to these concerns than the firm-level results for winners, and this qualification should be borne in mind throughout.

The dependent variable INNOVATION is the natural logarithm of one plus authorized invention patents, and SPILLOVERIN is the natural logarithm of one plus a non-winner's citations to award-winning firms' patents. Column (1) reports a SPILLOVERIN coefficient of 0.112 ($p < 0.01$), and Column (2) with full controls yields 0.088 ($p < 0.01$); in economic magnitude, a one-unit increase in SPILLOVERIN is associated with approximately 9.2% more authorized invention patents, supporting H2. RD enters at 0.542, confirming that non-winners investing more in R&D are better positioned to translate external knowledge into patentable inventions, and AGE at 0.028 suggests more established firms absorb and commercialize external flows somewhat more capably. A potential concern is that SPILLOVERIN may capture technological overlap rather than genuine absorption. Column (3) therefore adds PROXIMITYW, the mean Jaccard similarity between a non-winner's IPC codes and those of award-winning firms in the same year. The SPILLOVERIN coefficient declines modestly to 0.082 but remains significant at the 1% level, while PROXIMITYW enters at 0.124 ($p < 0.05$), indicating that technological proximity and active citation behavior contribute independently to innovation output. The persistence of the association after controlling for overlap is consistent with cross-organizational learning rather than a mechanical artifact of co-location in technology space, though it does not on its own rule out that unobserved innovativeness drives both citation behavior and patent output.

5. Discussion

This study finds that innovation honorary awards are positively associated with knowledge spillovers from award-winning firms, adding to work on how non-economic instruments shape diffusion. Where prior research has traced knowledge flows mainly through intellectual property regimes and government economic interventions (De Rassenfosse, Pellegrino, and Raiteri 2024), the findings extend that logic to honorary certification: a national award publicly validating a firm's technological leadership strengthens its appropriability position and lowers the imitation risk that usually discourages disclosure. The boundary conditions fit this account, as technological proximity and technology-intensive settings, where absorption is easier and citation networks denser, amplify the association (Battke et al. 2016; Li, Wei, and Wang 2015), while intense competition weakens it, stronger intellectual property protection strengthens it (Acs and Sanders 2012), and greater media visibility adds a reputational layer shaping how actively winners share.

The channel evidence speaks to how recognition connects to diffusion: award receipt is followed by more co-filing and more outward licensing, each tied to subsequent citation flows, though together they carry only part of the association and leave the remainder to channels not separately observable in the present data. The evidence on non-winners connects to work showing that firms learn from institutional peers (Cheng et al. 2021), as publicly codified knowledge flowing through citations is associated with higher invention output among non-winning firms even at arm's length. Tracing the chain from award receipt through winners' spillovers to non-winners' innovation gives firm-level grounding to the social rate-of-return argument that has long motivated innovation policy (Lucking, Bloom, and Van Reenen 2019), and suggests that recognition systems generate benefits reaching beyond the honored firm.

6. Conclusion

This study asks whether innovation honorary awards promote knowledge spillovers from winning firms and whether those spillovers benefit non-winning firms. Receiving the China Patent Award is associated with patent citation levels roughly 8% to 13% higher on average, concentrated in outward diffusion rather than internal recycling, and a within-firm comparison of awarded and non-awarded patents yields a smaller but still positive increment that, together with the surrounding checks, narrows the leading alternative explanations without amounting to a clean attribution. The association strengthens where technological bases are similar, where technology moves quickly and citation networks are dense, and under robust intellectual property protection and greater media visibility, while intense competition weakens it. Collaboration and outward licensing after award receipt both relate positively to later diffusion, though they carry only part of the association. More consequentially, knowledge absorbed from winners relates to roughly 9.2% more authorized invention patents among non-winners, holding technological proximity constant. The value of these awards thus lies not only in the direct incentive to recipients but in their tie to broader industrial progress.

These findings carry implications for policy design. Honorary awards work best as complements to fiscal subsidies and tax incentives, coordinating economic and non-economic instruments. The stronger diffusion among technologically similar firms suggests pairing award ceremonies with industry technology exchange platforms to help followers absorb frontier knowledge, and in intensely competitive industries, collaborative R&D incentives or patent alliances may offset the pressures that otherwise discourage sharing. Because licensing carries part of the diffusion, lowering the transaction costs of technology licensing, for instance through standardized platforms tied to recognized patents, could reinforce the system's reach. Continued strengthening of regional intellectual property protection lets winners share with greater assurance about their competitive position, and a higher public profile for award events may activate the reputational incentives that encourage sharing. For countries building or refining national innovation honors, the social returns from diffusion deserve as much weight as the direct incentives to recipients, and the effectiveness of such systems will depend on the surrounding market structure and intellectual property environment.

This study is subject to several limitations that also indicate directions for further research. Knowledge spillovers are measured through patent citations, which capture codified flows but not tacit knowledge moving through personnel mobility, informal contact, or industry conferences; future work drawing on inventor mobility records, co-authorship networks, or the text of licensing agreements could broaden the account of award-driven diffusion. The channel evidence is partial, as collaboration and licensing together carry about one fifth of the association, leaving the visibility of the award announcement, non-patent disclosure, and standard-setting to be distinguished in future work with finer data on post-award firm behavior. Because the award is conferred on merit rather than assigned by an external rule, the setting admits neither a sharp discontinuity nor a clean instrument, and the layered comparisons employed here, from firm-level matching through the within-firm design, narrow but do not remove the possibility that unobserved factors shape both award receipt and knowledge diffusion; the findings are accordingly best understood as robust associations consistent with a certification account, and a subsequent comparison against nominated-but-not-awarded patents, once such records become systematically available, or another source of sharp quasi-random variation, would help move these associations toward a stronger attribution. The downstream link for non-winning firms rests on firms' own citation behavior, which is itself a choice and therefore more exposed to reverse direction and unobserved innovativeness than the results for winners, making it the more tentative of the two links and a natural target for designs that separate absorption from selection. Finally, citation records include references added by patent examiners as well as by applicants, and if award recognition raises winning patents' visibility in examiners' prior-art searches, part of the observed citation increase may arise within the examination process itself. Two features of the evidence ease this concern: the acceleration in citations emerges only after the award rather than extending a pre-existing visibility trajectory, and citations to winners predict non-winners' subsequent patent output, a pattern more consistent with substantive knowledge use than with procedural referencing. As citation-source information for Chinese patents becomes more systematically available, separating applicant-added from examiner-added citations offers a straightforward way to sharpen this interpretation. The consistency of the association across the firm-level and patent-level analyses and across the several robustness checks nonetheless lends confidence in the pattern documented here.

Author contributions

CRediT: **Xue Lei:** Conceptualization, Formal analysis, Methodology, Writing – original draft; **Yuying Luo:** Data curation, Investigation, Writing – review & editing; **Xueguo Xu:** Supervision, Validation, Writing – review & editing.

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No potential conflict of interest was reported by the author(s).

Data availability statement

The data presented in this study are available on request from the corresponding author due to third-party restrictions.

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Appendix A. Additional robustness tests

A.1.#Propensity score matching

Using all control variables as covariates, 1:4 nearest neighbor matching produces a matched sample in which standardized biases for all covariates fall below 5%. Column (1) of Table A1 reports the matched-sample estimate: the AWARD coefficient is 0.118, significant at the 1% level.

A.2.#Dynamic panel estimation

The system GMM specification in Column (2) of Table A1 incorporates a lagged dependent variable and uses instruments lagged two or more periods. The AR (2) *p*-value of 0.342 and Hansen *p*-value of 0.268 support instrument validity, and the AWARD coefficient of 0.098 remains significant at the 5% level.

A.3.#Staggered adoption estimator

To address potential heterogeneity bias in the standard two-way fixed effects estimator arising from staggered treatment timing, Column (3) of Table A1 employs the Callaway and Sant'Anna (2021) estimator. The aggregated ATT of 0.112, significant at the 5% level, is modestly lower than the baseline TWFE estimate but remains economically meaningful.

Table A1. Matching, dynamic panel, and staggered adoption estimates.

Variable	(1) SPILLOVER PSM	(2) SPILLOVER System GMM	(3) SPILLOVER Staggered DID
AWARD	0.118***	0.098**	
	(2.76)	(2.04)	
ATT			0.112**
			(2.31)
L.SPILLOVER		0.468***	
		(19.48)	
Controls	Yes	Yes	Yes
Firm FE	Yes		
Year FE	Yes	Yes	Yes
N	19,824	18,942	24,236
Adj. R^2	0.418		
AR (2) *p*-value		0.342	
Hansen *p*-value		0.268	

Note: t-values in parentheses; ***, **, * denote significance at the 1%, 5%, 10% levels respectively.

A.4.#Placebo test

Award timing is randomly reassigned within the sample period and regression estimation is repeated 500 times. Figure A1 shows the resulting coefficient distribution centered around zero, while the actual quality-adjusted coefficient of 0.122 falls clearly in the right tail, indicating the baseline result is unlikely to be attributable to chance.

A.5.#Prior citation trajectory

Column (1) of Table A2 controls for PRECITATION, the mean annual citation count over the three years before the award year. The AWARD coefficient is 0.094 at the 5% level and PRECITATION enters at 0.186 at the 1% level, confirming that prior citation levels predict subsequent citations but that the award carries an additional association beyond this continuation. Column (2) uses ΔSPILLOVER as the dependent variable; the AWARD coefficient of 0.064, significant at the 5% level, is consistent with the post-award acceleration documented in Figure 2 rather than a level shift driven by pre-existing trends.

Table A2. Prior trajectory and growth-rate specifications.

Variable	(1) SPILLOVER	(2) ΔSPILLOVER
AWARD	0.094**	0.064**

(*Continued*)

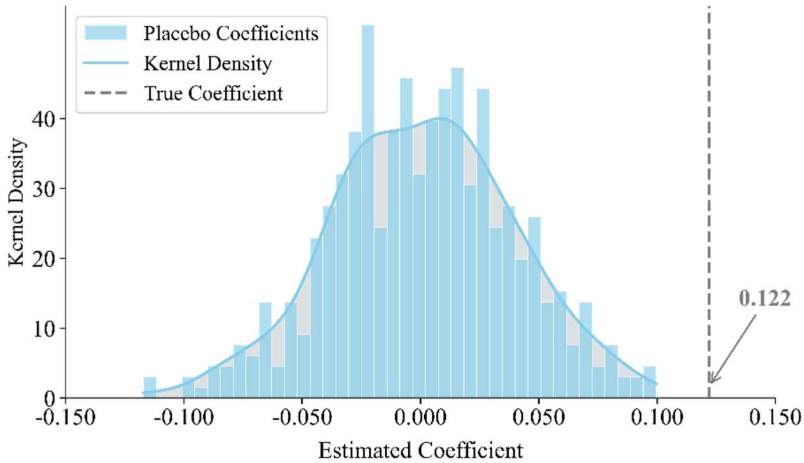

Figure A1. Placebo test.

Table A2. Continued.

Variable	(1) SPILLOVER	(2) ΔSPILLOVER
	(2.18)	(2.08)
PRECITATION	0.186***	
	(12.46)	
Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
N	22,486	23,648
Adj. R^2	0.448	0.284

Note: t-values in parentheses; ***, **, * denote significance at the 1%, 5%, 10% levels respectively.